Peer Response
by Abdulla Husain Salem Hadna Almessabi - Thursday, 7 August 2025, 1:08 PM
Hi Fahad,
I enjoyed the thorough details of how ABS has evolved and how it can be used. You have pointed out well the evolution of the monolithic to decentralised and flexible software structures, and I believe that the change has been necessitated by the non-linearity and complexities of present-day organisational ecosystems. The focus on support of autonomous agents in terms of real-time reactivity and scalability certainly is a major strength of ABS, and this growing importance is happening in dynamic settings such as financial markets and supply chains (Aghababaei & Koliou, 2023).

Emergent behaviour in ABS to micro-strategise and foresee macro tendencies is critical in the financial markets, which by their nature, are both complex and volatile (Haj Qasem et al., 2023). Besides, the perspective of distributed cognition in ABS, regarding decentralisation of decision making, poses the intriguing possibility for domains such as smart grids and autonomous logistics (Lange et al., 2021). ABS is not only a gain in technology but also a strategic decision to establish more autonomous and stronger systems (Onggo & Foramitti, 2021).

You have also talked about the greater organisational advantages of ABS, especially strategic foresight and innovation. My take is that ABS will be a critical tool in the move towards establishing self-governing digital ecosystems, as organisations will have to undergo high-fidelity simulations and adaptive processes. It is an invigorating possibility as companies can remain competitive and up to date with this constantly changing market with the help of the real-time learning abilities of agents (Zhang et al., 2021).

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by Abdulla Husain Salem Hadna Almessabi - Thursday, 7 August 2025, 1:10 PM

Hi Ali,
The post is a perfect introduction to ABS and its importance in a decentralised dynamic environment. I specifically like how you point out how ABS, rather than the classic systems, utilise the ability of independent agents who can not only perceive, make decisions, and interact to pursue self- or group objectives. Indeed, such flexibility is one of the fundamental strengths of the visual tool because individual organisations can work with a lot more elasticity and capacity (dos Santos & Saraiva, 2021).
Improved fault tolerance and decentralised decision-making are essential to industries that have to respond to the changes in real-time and be resilient, such as supply chain management, robotics, and healthcare (Fischbach et al., 2021). I believe that what you said about ABS supporting fault tolerance is particularly important in the present systems, where it is paramount to business continuity to ensure the system goes on regardless of the malfunctions that hit its agents. Further, its adaptability and learning capacities over time via machine learning, logical rules, and behavioural models are additional elements of sophistication that are traditionally missing in other systems (Tour et al., 2021).
You mentioned various models of agents, including reactive, deliberative, and hybrid models. The realisation that ABS can be used to facilitate cooperation among the agents depicts real-world interactions and thus the models are relevant in modelling activities of complex socio-economic systems (Turgut & Bozdag, 2023). They are used in financial markets, and your mention of them gives us useful information about market behaviours and patterns.
ABS can provide a useful framework under which the organisation would be able to make operations more efficient and achieve a strategic advantage due to decentralised decisions and versatility (Værbak et al., 2021).

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